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A Synthetic Data

A.1 Visual Navigation

As depicted in 1, given a specific k , the process of generating a 2D navigation map is composed of 3 steps, which are instruction generation, instruction simulation, map rendering. In instruction generation step, we enumerate all possible instruction sets navigating from the starting point to the destination (e.g move up, then move right). During this step, only the direction of each instruction is determined, while the moving distance is undetermined until next step. In instruction simulation step, simulation is applied in the 2D coordinate system with origin (0, 0) as the starting point. To guarantee an unique answer in each navigation map, the moving distance of each instruction is dynamically calculated to avoid overlapping. Each time when an overlapping is detected, the moving distance of previous instruction will be increased by 1 unit recursively until overlapping is resolved. As the distance is determined, those corresponding points are added to the navigating path. After all instructions are completed, the final point is marked as the destination. In the map rendering step, the bounding box of those points is adopted and normalized to a 2D square grid. The starting point and destination are marked with dedicated squares, and cells along the path are marked by empty squares, while other untouched cells are marked by obstacle squares.

Algorithm 1: Navigation Map Generation

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Input :  $k$ 
Output :  $\mathcal{C}_{solution} = \{s_1, s_2, \dots, s_n\}$ ,  $\mathcal{C}_{textual\_map} = \{t_1, t_2, \dots, t_n\}$ ,
           $\mathcal{C}_{visual\_map} = \{v_1, v_2, \dots, v_n\}$ ; where  $n = 2^{k+1}$ 
1  $dirs \leftarrow [\text{up, left, down, right}]$ 
2  $instruction\_sets \leftarrow \text{gen\_instruction}(dirs, k)$ 
3 for  $dir\_instructs$  in  $instruction\_sets$  do
4    $cur\_pos, path\_points, moves \leftarrow \text{initialize}()$  // origin as the starting point
5   for  $direction$  in  $dir\_instructs$  do
6     // instruction simulation
7     if not  $\text{validate\_plan}(cur\_pos, direction, path\_points)$  then
8        $\text{increase\_previous\_move}(cur\_pos, moves, path\_points)$ 
9     end
10     $cur\_pos, path\_points \leftarrow \text{step\_forward}(cur\_pos, direction, path\_points)$ 
11     $moves \leftarrow moves \cup direction$ 
12  end
13   $s_i \leftarrow moves$ 
14   $t_i, v_i \leftarrow \text{render\_map}(\text{extract\_bounding\_box}(path\_points))$ 
15   $\mathcal{C}_{solution} \leftarrow \mathcal{C}_{solution} \cup s_i$ 
16   $\mathcal{C}_{textual\_map} \leftarrow \mathcal{C}_{textual\_map} \cup t_i$ 
17   $\mathcal{C}_{visual\_map} \leftarrow \mathcal{C}_{visual\_map} \cup v_i$ 
18 end

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Since the direction of each navigation instruction is alternating, there are $4 * 2^{k-1} = 2^{k+1}$ kinds of spatial configurations for a k -hop navigation map. During the implementation, we simplify the recursive implementation with an early quit when path overlapping could not be resolved within one iteration, the main consideration of which is the size of the map. So the number of generated map is slightly lower than 2^{k+1} as the navigating step k increases.

A.2 Visual Tiling

The data generation process comprises 3 stages, including configuration generation, question generation and polyomino rendering. In the configuration generation stage, to generate valid spatial configurations of a rectangle and the corresponding polyomino set, we convert a tiling problem to existing formalized problems. One of the problems is an exact cover problem leveraging dancing link algorithm [Knu00], which could be described as: given a matrix of 0s and 1s, find a set of rows containing exactly one 1 in each column. The conversion is to construct a matrix of 0s and 1s, each row of which represents a possible arrangement of placing a specific polyomino in a rectangle. As illustrated in Equation 9, given k polyomino pieces, and a rectangle of n units to be filled, the first k